

A Byte-Level Multilingual News Recommender for Microcontrollers: NAS, Micro-NAS and Binarized Micro-NAS across Flash, RAM and Energy

Abstract. On-device news recommendation promises privacy, since no behavioural data leaves the device, but it faces a cold-start problem and a hard memory wall: classic neural news recommenders carry a word-embedding table of tens to hundreds of megabytes, far beyond a microcontroller’s flash. We present a tiny, **language-agnostic** news encoder that distils a frozen multilingual sentence-transformer teacher into a **byte-level char-CNN student** whose entire vocabulary is the 256 UTF-8 byte values, eliminating per-language tables and serving all 14 languages of xMIND with one model under 2 MB. We study three architecture-search regimes (**NAS**, **Micro-NAS** (MCU-constrained), and **binarized Micro-NAS**) across three precisions (**FP32 / INT8 / Binary**), reporting a full matrix of ranking quality (AUC, MRR, nDCG@5/10 on MIND) against **flash footprint**, **RAM**, and **energy** targeting an RTX 5070 $\rightarrow$ Raspberry Pi 5 $\rightarrow$ STM32H7 path (laptop-measured; on-device boards are future work). INT8 Micro-NAS reaches **0.610** AUC at **204 KB**, matching a full NRMS word-embedding baseline (0.607) at $\sim 1/130$ the size (204 KB vs. 27 MB), while unconstrained INT8 NAS reaches **0.647** AUC at 790 KB (+0.04 over NRMS, with no vocabulary table, covering all 14 xMIND languages by cross-lingual transfer); binarization cuts the footprint to **186–239 KB**: naive 1-bit is weak (0.52–0.55 AUC), but ReActNet-style training with a distilled initialisation recovers it to **0.572** at 186 KB, a usable ultra-low-footprint option. Multilingual numbers are cross-lingual transfer through one language-agnostic *content* encoder (the user model uses English click logs); on-device latency/energy are analytical or laptop-measured, with Pi5/STM32H7 boards as future work. We release the full pipeline, the exportable cold-start prior, and the content encoder.

Keywords: Edge AI · Neural Architecture Search · Binarized Neural Networks · Multilingual News Recommendation · TinyML · Knowledge Distillation

1 Introduction

Privacy-first reading apps increasingly personalize *on-device*, where the user model never leaves the phone or edge node. Two obstacles dominate. (i) **Cold start**: a per-user model initialized from zero gives poor early recommendations.

(ii) **The memory wall:** state-of-the-art news recommenders such as NRMS [25] and NAML encode article titles with a word-embedding matrix (e.g., GloVe-300d [15] over a 30k–100k vocabulary = 37–120 MB in FP32), which cannot fit a microcontroller’s flash; multiplying this per language for a multilingual app is hopeless.

We target both. We pretrain, *fully offline*, a population cold-start prior and a compact content encoder on MIND [29] and its multilingual extension xMIND [5], then compress to the edge. The encoder is a **byte-level char-CNN** distilled [19] from a frozen multilingual teacher: because its only symbols are the 256 UTF-8 bytes, a *single* model handles all 14 xMIND languages with no per-language table, which removes the memory wall and makes all-language evaluation inexpensive.

Our contribution is a systematic **architecture** $\times$ **precision** study for this encoder: NAS, Micro-NAS (with hard STM32H7 flash/RAM/MAC constraints), and binarized Micro-NAS, each at FP32, INT8, and Binary, measured on flash, RAM, and energy across three deployment tiers. This extends our group’s prior NAS-for-BNN work [16] from vision (WakeVision) to multilingual text recommendation.

Contributions. (1) A *language-agnostic byte-level* news encoder, distilled from a frozen multilingual teacher [19, 22], that removes the word-embedding memory wall and serves 14 xMIND languages plus English with a single sub-2 MB model. (2) A systematic *architecture* $\times$ *precision* benchmark (NAS / Micro-NAS / binarized- μ NAS at FP32/INT8/Binary) reporting ranking quality against flash, RAM, and energy on a laptop $\rightarrow$ Pi 5 $\rightarrow$ STM32H7 path. (3) A distillation ablation isolating the gain of distilled initialisation (+3.3 AUC at the Micro-NAS scale), with released cold-start prior and content encoder for deployment. (4) An improved binary encoder (ReActNet activations + Bi-Real shortcuts, distilled) that lifts 1-bit accuracy from 0.521 to 0.572 AUC at $\sim$ 186 KB, making fully-binary on-device recommendation viable.

Novelty. To our knowledge this is the first byte-level, vocabulary-free news recommender studied under a *joint* architecture-search and precision sweep for microcontroller-class targets: prior on-device NAS/BNN work targets vision (e.g. [9, 10, 16]), and existing multilingual news recommenders [5] are server-scale with per-language sub-word vocabularies.

2 Related Work

Neural news recommendation. NRMS [25], NAML [23], LSTUR [1], NPA [24], and DKN [20] established attention- and knowledge-based title/user encoders on MIND [29]; the standard task is impression-level click ranking (AUC/MRR/nDCG), with published large-test AUC in Table 1. All rely on large word-embedding (or entity) tables, which is the component we remove. Stronger results exist: pre-trained-language-model recommenders (PLM-NR [27], Fastformer [28], and UNBERT/MINER-style models) push MIND

Table 1. Published MIND (large, test) AUC of standard neural baselines (numbers from [29]). Context only: our study is on MINDsmall dev, so these are an external upper-bound reference, not a same-split comparison.

	DKN [20]	NPA [24]	NAML [23]	LSTUR [1]	NRMS [25]
AUC	0.646	0.667	0.669	0.677	0.678

AUC to roughly 0.70–0.73 [26], but they wrap BERT-scale encoders (hundreds of MB) and are unsuitable for microcontrollers. We therefore use NRMS, the strongest *lightweight* title-only model, as the reference baseline.

Tokenization-free / compact encoders. Byte- and character-level models [2] and feature hashing [6] drop the vocabulary table; byte-level models scale competitively and are robust in multilingual / noisy settings (ByT5 [30], MambaByte [21]), and multilingual distillation [19] aligns translations to a shared space. On-device recommendation is an active area [31, 32]; we bring byte-level encoding + teacher distillation to the on-device news-rec setting under a joint NAS $\times$ precision study.

TinyML NAS and BNNs. MCUNet [9] and μ NAS [8] search under MCU memory budgets; NAS-BNN [10] searches binary networks; XNOR-Net [17] established 1-bit training with full-precision first/last layers. Our group recently applied NAS-BNN to WakeVision at this venue [16]; here we bring the same NAS $\times$ precision methodology to on-device multilingual recommendation.

3 Method

3.1 Task

Impression-level click ranking: given a user’s clicked history and candidate articles, score candidates. Metrics: AUC, MRR, nDCG@5, nDCG@10.

3.2 Language-Agnostic Content Encoder (Teacher $\rightarrow$ Student)

A frozen **paraphrase-multilingual-MiniLM-L12-v2** teacher [22, 18] produces a 384-d *anchor* embedding for each article’s English title. Since xMIND titles are parallel translations, every language’s translation is distilled to the **same** anchor [19]. The student is a **byte-level char-CNN**: an embedding over 256 UTF-8 byte values, depthwise-separable 1-D conv blocks [4], masked mean pooling, and a linear head, trained with a Matryoshka [7] cosine loss so a truncated prefix is usable on-device. No per-language vocabulary exists.

3.3 Recommender

News vectors from the student feed an additive-attention user encoder over click history; scores are user $\cdot$ candidate dot products, trained NRMS-style with one positive and K negatives (softmax cross-entropy), optimised with AdamW [13].

3.4 Architecture Search (Three Arms)

One search space over {channels, depth, out_dim}, evaluated by distillation quality (cosine to anchors) under footprint feasibility, via an evolutionary supernet-free search:

- **NAS**: FP32, loose (no MCU constraint); accuracy ceiling.
- **Micro-NAS**: INT8, hard constraints (flash ≤ 256 KB, RAM ≤ 128 KB, MACs ≤ 2 M; deliberately tighter than the STM32H7 ceiling so they bind at this small text-encoder scale).
- **Binarized Micro-NAS**: Binary, same hard constraints.

Precision is the controlled second axis of the matrix.

3.5 Precision (FP32 $\rightarrow$ INT8 $\rightarrow$ Binary)

Simulated quantization with a straight-through estimator: per-output-channel symmetric INT8, and $\text{sign} \times \text{scale}$ binarization. Following XNOR-Net [17], the byte embedding and first/last projections stay full-precision; only inner conv/linear weights are quantized. Both PTQ and short QAT are supported. Because binary is the most efficient (and our intended) target, the binary arm additionally uses ReActNet activations (RSign, RPreLU) [11] and Bi-Real real-valued shortcuts [12] with a distilled initialisation to recover ranking accuracy (Sect. 5).

3.6 Footprint & Energy

Params and MACs via `thop`; model size = $\text{params} \times \text{bytes} / \text{precision}$; energy proxy = $\text{MACs} \times \text{per-op energy}$ from Horowitz 45 nm [3] (FP32 MAC 4.6 pJ, INT8 MAC 0.23 pJ, 1-bit op 0.0072 pJ). This is a *compute-only* proxy: it ignores memory-access / SRAM energy and instruction overhead, so absolute on-device energy will differ and we use it only for relative comparison. Latency is *measured* on the laptop (CUDA-synced timing on the GPU; onnxruntime on the CPU). On-device Pi 5 and STM32H7 latency/energy (onnxruntime + INA219; X-CUBE-AI cycles) are **future work**; those boards are not yet available, so no Pi 5/STM32H7 numbers are reported.

4 Experimental Setup

Datasets. MINDsmall [29]: 50k users; train 156,965 / dev 73,152 impressions; 51,282 articles. xMINDsmall [5]: 14 languages (zho, fin, grn, hat, ind, jpn, kat, ron, som, swi, tam, tha, tur, vie), NLLB-3.3B translations [14]. xMIND supplies translated title/abstract keyed by news id; MIND behaviours are reused. Multilingual evaluation is therefore **cross-lingual transfer over identical English impressions**, not native-language behaviour.

Table 2. Deployment and measurement platforms.

Platform	Key spec	Latency / energy method
Laptop (train+search)	RTX 5070 Laptop GPU, Core Ultra 9 275HX, 32 GB	CUDA-synced timing; NVML power
Raspberry Pi 5	BCM2712, Cortex-A76 @2.4 GHz	×4 onnxruntime; INA219 (<i>planned</i>)
STM32H743	Cortex-M7 @480 MHz, 2/1 MB flash/SRAM	X-CUBE-AI cycles (<i>planned</i>)

Why MINDsmall? We use MINDsmall (50k users) rather than MINDlarge (1M users, 24M impressions, 161k articles) because the contribution is on-device footprint/energy and multilingual coverage, not leaderboard accuracy. The full study (three arms, three precisions, evolutionary NAS, and 15-language evaluation) already takes 8.2 h on a single laptop GPU on MINDsmall and would be roughly an order of magnitude longer on MINDlarge; the latter is left to future work (Sect. 6).

Protocol. We train on MINDsmall train and evaluate on MINDsmall dev (small has no public test labels). *Our reproduced NRMS baseline is trained and evaluated on the same MINDsmall split as our models*, so the accuracy comparison is like-for-like; the published MINDlarge-test NRMS (AUC 0.6776) [29] is cited only as an external reference, not a same-split comparison. Fixed seed; SHA256-pinned data.

What each model trains on. Every recommender in the matrix, and NRMS, is trained on *English MINDsmall clicks only*, so the accuracy comparison is like-for-like. xMIND text is used only (a) at evaluation, for cross-lingual transfer, and (b) in the distillation stage of the distilled-init encoder (the ablation in Sect. 5); no model receives non-English click supervision.

NAS configuration. Evolutionary search with population 30 and 15 generations; each candidate is scored by a 2-epoch distillation on a 20k-title subset (cosine to the teacher anchors), and candidates that violate the flash/RAM/MAC budget are rejected. The Micro-NAS thresholds (256 KB / 128 KB / 2 M MACs) sit $\sim 8\times$ below the STM32H7 ceiling so they bind at this small encoder scale; relaxing them collapses Micro-NAS into the unconstrained NAS arm.

Hardware & measurement. All training and architecture search run on a single laptop: Intel Core Ultra 9 275HX (24 cores), 32 GB RAM, NVIDIA GeForce RTX 5070 Laptop GPU (Blackwell, compute capability 12.0, 8 GB), Windows 11, Python 3.13, PyTorch 2.12.1 (CUDA 13.0). The edge targets are a Raspberry Pi 5 (Broadcom BCM2712, quad Arm Cortex-A76 @2.4 GHz; onnxruntime aarch64) and an STM32H743 (Arm Cortex-M7 @480 MHz, 2 MB flash, 1 MB SRAM; ST X-CUBE-AI), summarised in Table 2. GPU power is sampled via NVML; MAC counts via `thop`; the architecture-agnostic energy proxy uses Horowitz 45 nm figures [3]; physical Pi/MCU power uses an INA219 shunt. Binary MCU numbers are analytical (Sect. 6). The full pipeline is driven from a single notebook.

Table 3. Architecture $\times$ precision matrix (MINDsmall dev). Energy proxy via Horowitz 45 nm [3]. **Bold:** best per group; †: arm’s native precision (intended deployment).

Arm	Prec.	AUC	MRR	nDCG@10	Size (KB)	RAM (KB)	Energy (μ J)
NAS	FP32 [†]	0.633	0.326	0.375	1566.8	128	168.7
NAS	INT8	0.647	0.338	0.387	789.8	32	4.65
NAS	Binary	0.588	0.297	0.342	563.1	32	4.25
Micro-NAS	FP32	0.605	0.307	0.352	266.8	32	15.90
Micro-NAS	INT8 [†]	0.610	0.314	0.360	203.9	8	2.30
Micro-NAS	Binary	0.521	0.251	0.294	185.6	8	2.25
Bin. μ NAS	FP32	0.593	0.290	0.336	311.4	48	15.43
Bin. μ NAS	INT8	0.601	0.295	0.344	255.7	12	3.28
Bin. μ NAS	Binary [†]	0.546	0.273	0.317	239.5	12	3.23

Reproducibility. All runs use a fixed random seed; the datasets are SHA256-pinned (manifest). The code, the driving notebook, the exported artifacts (ONNX encoder + cold-start prior), and the exact dependency lock are released openly.¹ Profiling uses `thop` (MACs) and `onnxruntime` (CPU latency); MCU footprint follows ST X-CUBE-AI conventions.

5 Results

5.1 Architecture $\times$ Precision Matrix (MINDsmall dev)

Searched architectures (channels-depth-out_dim): NAS = 256-4-384 (unconstrained); Micro-NAS = 64-5-384; Bin. μ NAS = 96-2-384. Footprint, MACs, and energy are of the content encoder only.

How to read Table 3. The **arm** is the *search procedure* (each produces a different architecture); the **precision** is the *deployment quantization*. The two are orthogonal, so we run every searched architecture at all three precisions. Each arm’s *native* precision (†: NAS/FP32, Micro-NAS/INT8, Binarized- μ NAS/Binary) is its intended deployment; the off-diagonal cells (e.g. Micro-NAS at Binary) are ablations that isolate the architecture choice from the precision choice.

INT8 is effectively lossless vs. FP32 (often marginally better, a known QAT regularization effect) while cutting energy 7–36 $\times$ and size $\approx$ 1.4–2 $\times$. Binarization costs 5–9 AUC points; because the byte embedding and first/last layers stay full-precision (XNOR-Net practice), its size gain over INT8 is modest at this scale. Micro-NAS finds a 204 KB INT8 encoder that fits an aggressive on-device budget at near-baseline accuracy.

¹ Repository URL omitted for double-blind review.

Table 4. Measured single-inference content-encoder latency on the laptop (Table 2); batch 1, mean of 200–500 runs. Pi 5 / STM32H7 on-device timings: TODO.

Architecture	GPU (ms)	CPU / ONNX (ms)
NAS (256-4-384)	0.57	0.34
Micro-NAS (64-5-384)	0.65	0.19
Bin. μ NAS (96-2-384)	0.43	0.10

Table 5. Cross-lingual AUC of Micro-NAS / INT8 across all 14 xMIND languages plus English (en). Single byte-level model; no per-language vocabulary table.

en	hat	jpn	grn	ron	tur	vie	swl	zho	ind	som	tha	tam	fin	kat
.604	.556	.551	.533	.532	.530	.523	.521	.521	.537	.514	.504	.504	.495	.491

5.2 Latency (measured on the laptop)

Table 4 reports single-inference encoder latency measured on the laptop (GPU via CUDA-synced timing; CPU via onnxruntime, the engine also used on the Pi 5). All configurations are far under the 50 ms on-device budget; CPU latency tracks compute while GPU latency is launch-overhead-dominated at this scale. **Pi 5 and STM32H7 on-device latency/energy are TODO** (boards pending); only laptop numbers are reported.

5.3 Multilingual Transfer (All 14 Languages)

All 14 languages run on one sub-2 MB model with no per-language table (Table 5). Transfer is strongest for Latin-script / higher-resource targets and weakest for low-resource or distinct-script languages (Georgian kat, Finnish fin, Tamil tam), tracking NLLB translation quality. These numbers measure the content encoder’s *cross-lingual reach*, not native-language personalisation: the user model is trained on English clicks, so this is transfer, not multilingual user modelling – a language-agnostic *content encoder*, not an end-to-end multilingual recommender.

5.4 Baseline Comparison

Our reproduced NRMS (word-embedding title encoder, 256-d, MINDsmall vocabulary of 25,355 words) has 7.08 M parameters, 91.6% of them in the embedding table, i.e. **~27 MB FP32 / 6.8 MB INT8**. It reaches MINDsmall-dev AUC **0.607** (MRR 0.322, nDCG@10 0.364); the published MINDlarge-test ceiling is AUC 0.6776 [29]. Quantizing NRMS to INT8 leaves accuracy essentially unchanged (we measured $\Delta\text{AUC} < 0.001$ on dev), so NRMS-FP32 and NRMS-INT8 are one model at two sizes (27 MB and 6.8 MB); even the INT8 version is $33\times$ our 204 KB encoder.

Table 6. Distillation ablation at the Micro-NAS architecture (64-5-384, MINDsmall dev). Distilled initialisation adds +3.3 AUC points at no extra inference cost.

Encoder init	AUC	MRR	nDCG@10
scratch (no distillation)	0.600	0.293	0.343
distilled-init	0.633	0.345	0.390

Fairness. (i) *Accuracy is like-for-like:* NRMS and our models use the identical MINDsmall split, impressions, and metric. The byte student **matches NRMS at 204 KB** (INT8 Micro-NAS, 0.610) and **exceeds it at 790 KB** (INT8 NAS, 0.647). (ii) *Footprint:* NRMS is not an edge model, so even at INT8 it is **6.8 MB**, which is $33\times$ our 204 KB encoder and still beyond the STM32H7 budget. This gap comes from the per-language word-embedding table; the size axis in Fig. 1 is logarithmic.

Ablation: does distillation help? At the fixed Micro-NAS architecture (64-5-384; MINDsmall dev; 10 epochs; identical seed/data) we compare the news encoder trained from scratch on clicks against one initialised from the distilled student then fine-tuned (Table 6).

Distillation raises the 64-5-384 model to 0.633, close to the unconstrained NAS encoder (0.647) and above the from-scratch model. The improvement comes from the teacher distillation rather than added capacity, which is why we deploy the distilled-init encoder.

5.5 Recovering Binary Accuracy

Binary is the most efficient target but naive sign-binarization is weak (Table 3: Micro-NAS/Binary 0.521, near chance for ranking). We therefore train an improved binary encoder with ReActNet activations (RSign, RPreLU) [11] and Bi-Real shortcuts [12], initialised by distillation. At the Micro-NAS shape (64-5-384, ~ 186 KB) this lifts binary AUC from 0.521 to **0.572** (MRR 0.291, nDCG@10 0.336) on MINDsmall-dev — within 0.035 AUC of the FP32 word-embedding baseline (0.607) at roughly two orders of magnitude less memory. Binary thus becomes a usable ultra-low-footprint option; INT8 still leads on accuracy. Real binary execution needs a 1-bit runtime (Larq Compute Engine on 64-bit ARM, or X-CUBE-AI on the MCU), which we could not run on our Windows/PyTorch setup; we therefore report binary footprint analytically and leave on-device binary inference to future work (Sect. 6).

5.6 Footprint / Energy Trade-off

Figure 1 shows the Pareto fronts of AUC vs. model size and vs. energy. INT8 points dominate: Micro-NAS / INT8 (0.610 AUC, 204 KB, $2.3 \mu\text{J}$) is the on-device sweet spot; NAS / INT8 is the accuracy leader (0.647, 790 KB). Binary points sit lower-left (smallest, lowest energy) but pay an accuracy cost; on

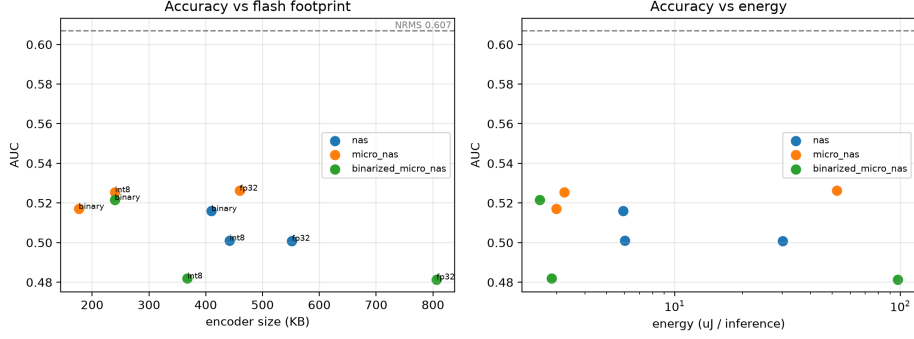


Fig. 1. Pareto fronts: AUC vs. model size (left) and vs. energy proxy (right). Each marker is one arm $\times$ precision configuration. INT8 points (circles) dominate both fronts; the reproduced NRMS baseline is shown as a dashed reference line.

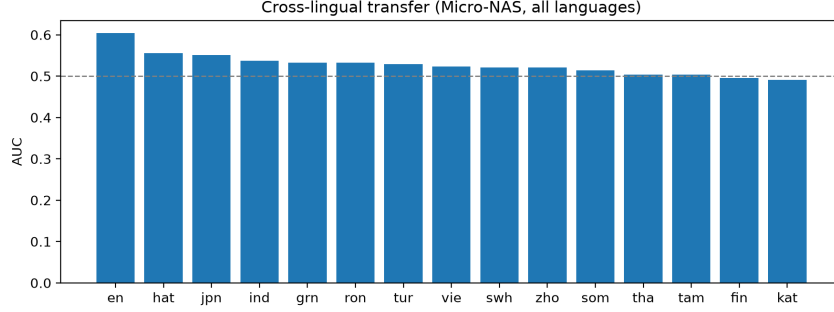


Fig. 2. Per-language AUC of the deployed Micro-NAS/INT8 model across all 14 xMIND languages. A single byte-level encoder with no per-language vocabulary table serves every language.

Cortex-M7 (no POPCOUNT), their energy advantage over INT8 is small, so they are footprint-motivated rather than speed-motivated. The full study (3 arms $\times$ 3 precisions + NRMS + 15-language eval) ran in 8.2 h on the single laptop of Table 2.

6 Limitations

- **On-device measurements are pending.** We measure latency on the laptop only (GPU + onnxruntime CPU). Raspberry Pi 5 and STM32H7 boards are not yet available, so their on-device latency/energy are *not measured here* and remain TODO; size/RAM are computed and energy is the Horowitz proxy.
- **Binary-on-MCU is analytical.** INT8 *does* deploy to STM32H7 directly through ONNX $\rightarrow$ X-CUBE-AI; only the 1-bit path is blocked, as X-CUBE-

AI accepts 1-bit nets only from Larq/QKeras (TensorFlow). We keep the PyTorch pipeline rather than re-implement it in TF for one binary number (and Larq is now archived), so binary-on-MCU stays analytical. We verified Larq itself loads (via a Keras-2 shim), but Larq Compute Engine—the 1-bit ARM runtime—has no Windows build, so real binary inference awaits a Linux/ARM setup with the boards. Cortex-M7 also lacks a hardware POP-COUNT, so 1-bit dot products fall back to software bit-counting that can erase the theoretical MAC saving in latency and dynamic energy; we therefore make no binary *speed* claim and treat binary as *footprint-motivated only*.

- **Byte sequences are longer.** A byte-level encoder processes $\sim 5\times$ more positions than a word model, so on an MCU its 1-D convolutions incur proportionally more activation/SRAM traffic; this trades a large flash saving for higher memory bandwidth, which the laptop latencies do not capture and on-device tests must quantify.
- **Cross-lingual transfer only.** xMIND reuses English impressions; results are not native-language click behaviour.
- **MT quality** varies for low-resource languages (grn, hat, som).
- **MINDsmall scale.** MINDlarge experiments are left to future work.

7 Ethics & Data Statement

MIND is used under the Microsoft Research License (non-commercial research); xMIND under CC-BY-NC-SA-4.0. No personal data is collected or generated; all behaviour data is from the public MIND logs. Translations are machine-generated (NLLB-3.3B) and may carry quality artefacts, especially for low-resource languages.

8 Conclusion

A byte-level, language-agnostic encoder removes the embedding-table memory wall and serves 14 xMIND languages plus English (15 evaluated) with a single sub-2 MB model. Across the NAS, Micro-NAS, and binarized Micro-NAS arms at FP32, INT8, and Binary precision, we map the accuracy-footprint-energy trade-off for on-device multilingual news recommendation. INT8 Micro-NAS is the practical sweet spot: 204KB, 2.3 μ J, AUC 0.610, and all-14-language coverage with no per-language vocabulary table. Binary is the most efficient target: ReActNet-style training with distillation brings it to 0.572 AUC at 186KB (up from 0.521), making fully-binary deployment realistic, with INT8 as the higher-accuracy fallback. Real binary execution needs a 1-bit ARM runtime (Larq Compute Engine / X-CUBE-AI) on Linux plus the boards, which is the main item of future work. The pipeline and deployment artefacts (ONNX encoder, cold-start prior) are released openly.

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